DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2024

urse: B. Tech. **Branch :** Electronics Engineering and Allied

bject Code & Name: Probability Theory and Random Processes BTBS404 **Semester :** IV

ıx Marks: 60 **Date:** 27/12/2024 **Duration:** 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Objective type questions. (Compulsory Question) **12**

1 Which of the following is true about probability? CO1 1
- a) Probability is always greater than 1
- b) Probability of any event is always between 0 and 1
- c) Probability of a certain event is 0
- d) Probability of an impossible event is 1

2 Which of the following is an example of a dependent event? CO1 1
- a) Rolling a die twice
- b) Drawing a card from a deck and then drawing another without replacement
- c) Tossing a coin
- d) Choosing a random number from 1 to 10

3 Which of the following is an example of a discrete random variable? CO1 1
- a) Height of a person
- b) Number of heads in 10 coin tosses
- c) Time taken to complete a task
- d) Weight of a watermelon

4 The probability density function (PDF) for a continuous random variable satisfies: CO2 1
- a) It is always greater than 1
- b) Its integral over the entire sample space is 1
- c) It can be negative
- d) It is always equal to 1

5 If X and Y are two independent random variables, what is the expected value of X + Y? CO2 1
- a) $E(X + Y) = E(X) + E(Y)$
- b) $E(X + Y) = E(X) - E(Y)$
- c) $E(X + Y) = E(X) * E(Y)$
- d) $E(X + Y) = 0$

6	In the context of random vectors, what is linear independence? a) No component is a linear combination of the others b) All components have the same distribution c) All components have identical means d) The components are dependent on each other	CO2	1
7	The Central Limit Theorem (CLT) applies to: a) The sum of a large number of independent random variables b) The sample mean of a non-random sequence c) A fixed set of random variables d) Non-independent random variables	CO3	1
8	The correlation between two random variables X and Y is: a) A measure of their linear dependence b) Equal to the product of their means c) Always zero d) The square of their covariance	CO3	1
9	If two random variables X and Y are independent, then their covariance is: a) 1 b) 0 c) E(XY) d) E(X)E(Y)	CO3	1
10	The power spectral density of a random process describes: a) The distribution of the process' energy over time b) The frequency distribution of the process' power c) The variance of the random process d) The autocorrelation function of the process	CO4	1
11	Which of the following is true for white noise? a) It has constant mean and variance over time b) It has zero mean and constant variance c) It has a non-zero mean d) It is a deterministic sequence	CO4	1
12	The cross-correlation function of two processes describes: a) The similarity between two processes at different times b) The autocorrelation of each individual process c) The dependence of the processes on time d) The variance of each process	CO4	1

Q. 2	**Solve the following.**			**12**
	A)	State and explain Bayes' theorem with an example.	CO1	6
	B)	What are joint and conditional probabilities? Derive the formula for joint probability in terms of conditional probability.	CO1	6
Q.3	**Solve the following.**			**12**
	A)	Explain the concepts of probability mass function (PMF) and probability density function (PDF). How are they different? State their properties.	CO2	6
	B)	The probability distribution of a discrete random variable X is given as follows: $P(X = 1) = 0.3$, $P(X = 2) = 0.4$, $P(X = 3) = 0.3$. Find the expected value (mean), variance and standard deviation of X.	CO2	6
Q. 4	**Solve Any Two of the following.**			**12**
	A)	Describe the characteristics of the normal (Gaussian) distribution. What are its key properties?	CO2	6
	B)	i) Define marginal probability distribution and conditional probability distribution? ii) The joint distribution of two random variables X and Y is given by: $P(X = 1, Y = 1) = 0.2$, $P(X = 1, Y = 2) = 0.3$, $P(X = 2, Y = 1) = 0.5$. Find the marginal distribution of X and Y.	CO2	6
	C)	What is a function of random variable? How to find $f_y(y)$ when $f_x(x)$ is known?	CO3	6
Q.5	**Solve Any Two of the following.**			**12**
	A)	What is almost sure convergence? Explain with an example.	CO3	6
	B)	State and prove Tchebycheff Inequality.	CO3	6
	C)	Explain the Central Limit Theorem (CLT) and its significance in probability theory.	CO4	6
Q.6	**Solve Any Two of the following.**			**12**
	A)	Show that the random process defined as $X(t) = A \cos(w_0 t + \theta)$ is wide sense stationary, where θ is a random variable uniformly distributed between 0 and 2π. A and w_0 are constants.	CO4	6
	B)	Write a note on Ergodicity.	CO4	6
	C)	What is meant by strict-sense and wide-sense stationarity? How are they different?	CO4	6

***** End *****